



Interference analysis of anti micro and small unmanned aerial vehicles and mobile communication systems uplink

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Abstract

The micro and small unmanned aerial vehicles (msUAVs) internet of things has been widely used in different fields. At the same time, the disordered flight of msUAVs inevitably become a potential threat to public security (such as airports, railway stations, major event venues, etc.). In view of the safety problem of msUAVs, the anti-micro and small unmanned aerial vehicles (AmsUAVs) is used to solve the defense problem of msUAVs in public safety places. The system blocks the communication link of the msUAVs by transmitting high power to achieve the purpose of low altitude control. At the same time, the AmsUAVs will also cause interference to the uplink of other nearby radio communication systems, such as civil aviation and mobile communication systems. In this paper, an analysis method based on the interference-to-noise ratio (INR) criterion is proposed and interference analysis is performed on AmsUAVs and some mobile communication systems uplink. First, to control msUAVs, it is necessary to calculate the minimum transmit power of the AmsUAVs in the corresponding frequency band according to the uplink and the radio wave propagation distance of the msUAVs. Second, to avoid harmful interference to other radio communication systems (e.g., CDMA and TD-LTE), the frequency and distance separation between the AmsUAVs and mobile communication systems should be calculated according to factors such as application scenarios, radio wave propagation distance, and working frequency band. Finally, a simulation analysis provides the transmit power, off-axis angle, frequency separation, and distance separation of the AmsUAVs and mobile communication base station in different environments.

Keywords Anti micro and small unmanned aerial vehicles · Mobile communication base station · Interference analysis · Uplink · Interference-to-noise ratio criterion

1 Introduction

In recent years, micro and small unmanned aerial vehicles (msUAVs) has made rapid development due to its flexible flight mode and broad application prospects [1–3]. With low cost, small size, and light weight, msUAVs can perform remote operations conveniently, acquire high-resolution images, access hard-to-reach areas, and obtain

flexible response to complex geographical environments, the msUAVs internet of things has played an important role in rescue, disaster relief, aerial photography, unmanned mapping, agricultural plant protection, electric power inspection and other scenarios [4, 8]. Over time, the msUAVs internet of things will provide more accurate and convenient services in different fields, including surveillance, security, search and rescue missions, and environmental monitoring [9, 12].

However, the abuse of the msUAVs internet of things has a significant impact on public security [13]. Due to the portability and ease of operation of msUAVs, the use of consumer-grade msUAVs has increased explosively, and the problem of disordered flight has become increasingly serious. The corresponding potential threat of msUAVs flight to public safety has become increasingly serious. The phenomenon of random flight has greatly threatened aviation safety, railway safety, major event venue, security of

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sensitive areas, security of key engineering projects, and even some msUAVs are directly used for illegal and criminal activities [14, 15]. Incidents of msUAVs flight affecting the take-off and landing safety of civil aviation aircraft and threatening the security of important areas and facilities occur frequently. To reduce the existing or potential security problems of civil msUAVs to other areas and facilities, the control technology of msUAVs has emerged and has been widely used [16, 17].

An anti micro and small unmanned aerial vehicles (AmsUAVs) is a radio technology for civilian msUAVs control [18], which is closely related to the msUAVs radio communication system. The common AmsUAVs [19] is mainly the radio-based msUAVs control technology, which achieves control by blocking the communication link of the msUAVs, thus hindering its normal operation and forcing landing. Therefore, in general, the AmsUAVs aims to block one or more of the above communication links to divert or force the landing of civil msUAVs to realize the control.

Nevertheless, in the process of controlling the radio links of msUAVs, high power signal suppression is commonly adopted, which will cause interference not only to the communication link of msUAVs but also to other radio systems that use the same frequency band or adjacent frequency band, such as the GPS altitude measuring equipment on aircraft, GPS positioning equipment of mobile communication base stations, and the uplink and downlink of mobile communication systems. The high spread spectrum gain of the GPS signal results in poor performance of some frequency band interference of the AmsUAVs. Therefore, the AmsUAVs will normally control the uplink remote control link and downlink information transmission link of msUAVs.

In general, the msUAVs is easier to find than ground controller (GC) of msUAVs, and the radio transmission conditions between the AmsUAVs and msUAVs are better. Therefore, it is easier to control the msUAVs by suppressing its uplink. According to the report ITU-R M.2237 [23] and the frequency division of ITU region 3, as well as the frequency usage of the existing civil msUAVs, AmsUAVs may cause harmful interference to the base station of mobile communication in the direction of interfering signal transmission in the process of controlling msUAVs.

Therefore, this paper aims to conduct interference analysis and determine the minimum frequency and distance intervals required for both AmsUAVs and mobile communication systems to coexist. Thus, the study focuses on preventing the high power repressive interference generated by the AmsUAVs from affecting the uplink of base station.

The rest of this paper is organized as follows. The system model is presented in Sect. 2, and the proposed interference analysis of AmsUAVs and mobile communication systems is presented in Sect. 3. Numerical examples are carried out in Sect. 4 to verify the derived result. Finally in Sect. 5 we conclude the main results of this paper.

2 System model

We consider the interference between the AmsUAVs and mobile communication system as shown in Fig. 1 [24], where the AmsUAVs operates in the adjacent frequency band of mobile communication system. The msUAVs comprises a msUAV and a GC, which is a type of remote control equipment and usually fixed on the ground or handheld. The mobile communication system is equipped with a base station and mobile terminals. The AmsUAVs controls the msUAVs by high power noise interference on the uplink remote control link. In Fig. 1, Line 1 is the control link of the AmsUAVs to the msUAVs, Line 2 is the interference link of the AmsUAVs to the uplink of the base station, and Line 3 and Line 4 are the communication links of the msUAVs and mobile communication systems, respectively.

3 Interference analysis of AmsUAVs and mobile communication systems

Wireless interference always exists in the radio system [25, 26]. The power of the interference signal is closely related to the distance between the transmitter and the receiver. As the distance increases, the interference signal decreases. Whether one radio system is harmful to another is assessed by interference analysis, which is usually based on a simple relationship [27, 28]. A jamming signal with power within the allowable range of the receiver will

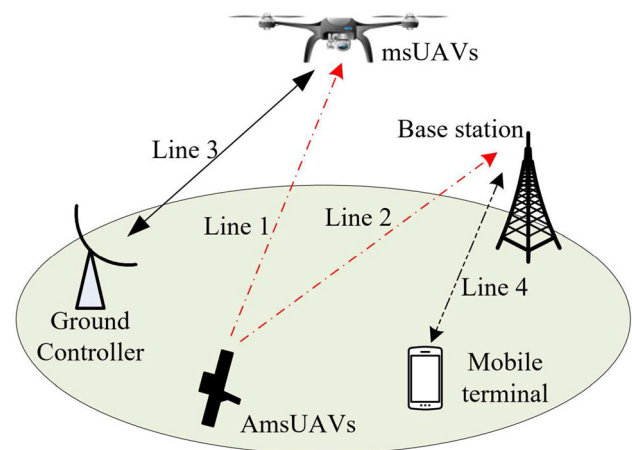


Fig. 1 System model

generally not cause harmful interference to the receiver [22]. In this paper, interference analysis of a wireless communication system is performed based on the interference-to-noise ratio (INR) criterion to determine the ranges of frequency and distance separation for the two systems to coexist in time, frequency, and space domains [29].

3.1 Interference-to-noise ratio criterion

Due to msUAVs generally use frequency hopping and spread spectrum technology, and frequency hopping parameters can be self-adaptive, they have a certain ability of anti-jamming. Generally, in order to drive away or forcibly land the msUAVs faster, high-power noise is used to cover all communication frequency bands of the msUAVs.

When the AmsUAVs controls the uplink of the msUAVs, the power of the transmitted signal, which is regarded as the noise by the msUAVs, must be increased to exceeds the INR protection requirements of the msUAVs, thereby allowing it to block the control of the GC over the msUAVs. The INR is defined as

$$\text{INR} = \frac{I}{N}, \quad (1)$$

where I and N denote the interference and noise power in W, respectively. The units of power can be converted into dB as $\text{INR}(\text{dB}) = I - N$.

According to report ITU-R M.2119 [30], the long-term protection requirement for a msUAVs is -3dB and the short-term protection requirement is 0dB . To avoid harmful interference to the mobile communication system of an adjacent channel, the power of the interference signal received by the base station must meet the INR protection requirements of normal communication quality of service. Such as, the INR protection requirement of the CDMA base station and TD-LTE base station are -7dB , that is $\eta_{\text{CDMA}} = \eta_{\text{TD-LTE}} < -7\text{dB}$. Thus the INR criterion can be defined as

$$\text{INR}(\text{dB}) = \begin{cases} I_{\text{msUAVs}} - N_{\text{msUAVs}} \geq \eta_{\text{msUAVs}}, & \text{msUAVs} \\ I_{\text{bs}} - N_{\text{bs}} < \eta_{\text{bs}}, & \text{base station} \end{cases}, \quad (2)$$

where I_{msUAVs} and N_{msUAVs} are the interference and noise power received by the msUAVs receiver, respectively. η_{msUAVs} is the msUAVs's INR protection requirement. For the AmsUAVs to block the msUAVs communication system, the requirement $\text{INR} \geq \eta_{\text{msUAVs}}$ must be met. I_{bs} and N_{bs} are, respectively, the interference and noise power received by the base station. To ensure the normal quality of service, the requirement $\text{INR} < \eta_{\text{bs}}$ (e.g., η_{CDMA} and $\eta_{\text{TD-LTE}}$) must be met.

3.2 Calculation of interference power

For msUAVs or GC, the received interference power is a function of the transmit power, antenna gain, path loss, etc.. The interference power received by the jammed radio receiver is expressed by

$$I = P_t + G_t - L_t + G_r - L_r - L_p(d) - FDR(\Delta f), \quad (3)$$

where, P_t is the transmit power of the interference source (dBm); G_t and G_r are the antenna gain of the transmitter and the receiver, respectively; L_t and L_r are the feeder loss of the AmsUAVs and base station, respectively; and $L_p(d)$ is the transmission path loss for a separation distance d between the interferer and receiver (dB). $FDR(\Delta f)$ is a frequency-dependent rejection of a certain frequency separation (dB), which is a measure of the rejection produced by the receiver selectivity curve on an unwanted transmitter emission spectra, defined as

$$FDR(\Delta f) = -10 \log \frac{\int_{-\infty}^{+\infty} P(f)(H(f + \Delta f))^2 df}{\int_{-\infty}^{+\infty} P(f) df} \text{ dB}, \quad (4)$$

where, f is the operating frequency; $P(f)$ is the power spectral density of the interfering signal equivalent intermediate frequency (IF); $H(f)$ is the frequency response of the receiver; Δf is the frequency separation between jammed receivers and interfering transmitters, defined as $\Delta f = f_i - f_r$, where f_i is the interferer-tuned frequency and f_r is the receiver-tuned frequency. In particular, $\Delta f = 0$ indicates co-channel interference.

As can be seen from (4), the interference signal power is $P_t = \int_{-\infty}^{+\infty} P(f) df$, $(H(f + \Delta f))^2$ reflects the characteristics of the receiver spectrum mask, and $FDR(\Delta f) \geq 0$.

It is evident from (4) that $FDR(\Delta f)$ is strongly dependent on the extent of overlapping between the receiver passband and the power spectrum of the interfering signal. As Δf increases, the extent of overlapping diminishes, thus resulting in lower interference power, and higher $FDR(\Delta f)$. Considering the effect of the off-axis angle on the antenna gain, the off-axis angle of the transmitting antenna is defined as θ_1 and that of the receiving antenna is defined as θ_2 . Then the interference power is

$$I = P_t + G_t(\theta_1) - L_t + G_r(\theta_2) - L_r - L_p(d) - FDR(\Delta f), \quad (5)$$

where, $G_t(\theta_1)$ is the antenna gain of the AmsUAVs in the direction of the receiver when the off-axis angle is θ_1 (dBi); $G_r(\theta_2)$ is the antenna gain of the jammed receiver (e.g., msUAVs) in the direction of interference when the off-axis angle is θ_2 (dBi).

According to the calculation method of a 100MHz–70GHz antenna radiation pattern given in ITU-R F.699-8 [31] and ITU-R F.1245-1 [32] recommendations, the antenna gain $G_{\Lambda}(\theta_i)$ ($\Lambda = t, r; i = 1, 2, 3$) corresponding to different off-axis angles θ_i is given as follows.

Assume that D is the diameter or length of the antenna and λ is the wavelength. When $D/\lambda > 100$,

$$G_{\Lambda}(\theta_i) = \begin{cases} G_{\max} - 2.5 \times 10^{-3} \left(\frac{D}{\lambda} \theta_i \right)^2, & 0^\circ < \theta_i < \varphi_m \\ G_1, & \varphi_m \leq \theta_i < \max(\varphi_m, \varphi_r), \\ 29 - 25 \log \theta_i, & \max(\varphi_m, \varphi_r) \leq \theta_i < 48^\circ \\ -13, & 48^\circ \leq \theta_i \leq 180^\circ \end{cases} \quad (6)$$

where $G_{\max}$ is the maximum gain for the antenna and G_1 is the first side lobe gain, θ_{3dB} is a 3dB beam width, $G_{\max} = 20 \log(D/\lambda) + 7.7$, $G_1 = 2 + 15 \log(D/\lambda)$, $\varphi_m = 20D\sqrt{G_{\max} - G_1}/\lambda$ and $\varphi_r = 12.02(D/\lambda)^{-0.6}$.

When $D/\lambda \leq 100$,

$$G_{\Lambda}(\theta_i) = \begin{cases} G_{\max} - 2.5 \times 10^{-3} \left(\frac{D}{\lambda} \theta_i \right), & 0^\circ < \theta_i < \varphi_m \\ 39 - 5 \log \left(\frac{D}{\lambda} \right) - 25 \log \theta_i, & \varphi_m \leq \theta_i < 48^\circ \\ -3 - 5 \log \left(\frac{D}{\lambda} \right), & 48^\circ \leq \theta_i \leq 180^\circ \end{cases} \quad (7)$$

3.3 Calculation of path loss

A spatial factor is used in the computation of distance-related signal attenuation; it is closely related to the radio propagation model adopted in this study and the statistical distribution of the interfering signal at the front end of the interfered receiver [33].

Path loss is calculated by radio propagation models. In this study, a radio propagation model is selected according to the system configuration, system bandwidth, operating frequency band, and geographical environment surrounding the service area [34].

Considering that the AmsUAVs generally uses high-power signal suppression, in order to minimize its effect on other surrounding radio services, directional antenna is used for jamming the msUAVs. To attain the minimum interference power of the AmsUAVs, the free-space radio propagation model is used to calculate the path loss by the following formula:

$$L_p(d) = 32.4 + 20 \log d + 20 \log f. \quad (8)$$

In the interference analysis of the AmsUAVs and mobile communication systems uplink, path loss calculation methods of urban, suburban, or open scenarios in the

Okumura–Hata model [35] are adopted according to different application scenarios. The formulas of path loss in the three scenarios are as follows.

Radio propagation model in the urban scenario is

$$L_p(d) = 49.55 + 26.16 \log f - 13.82 \log h_1 - \alpha(h_2) + (44.9 - 6.55 \log h_1) \log d, \quad (9)$$

where h_1 denotes the effective height of the base station antenna (m); h_2 denotes the effective height of the mobile station antenna (m); and $\alpha(h_2)$ represents the height factor of the mobile station antenna (dB), and when $f > 400$ MHz, it is defined as

$$\alpha(h_2) = 3.2[\log(11.75h_2)]^2 - 4.97, \quad (f > 400\text{MHz}). \quad (10)$$

Radio propagation model in the suburban scenario is

$$L_p(d) = 69.55 + 26.16 \log f - 13.82 \log h_1 - \alpha(h_2) + (44.9 - 6.55 \log h_1) \log d, \quad (11)$$

where $\alpha(h_2)$ is defined as

$$\alpha(h_2) = 2 \left[\log \left(\frac{f}{28} \right) \right]^2 + 5.4. \quad (12)$$

Radio propagation model in the open scenario is

$$L_p(d) = 28.6 + 26.16 \log f - 13.82 \log H_b + [44.9 - 6.55 \log(H_b)] \log d - 4.78(\log f)^2 + 18.33 \log f, \quad (13)$$

where H_b represents the height of the base station.

3.4 Calculation of minimum interference power

The transmit power of the AmsUAVs is closely related to the bandwidth, antenna gain, and anti-interference threshold of the receiver of the msUAVs. In practice, the maximum power received by the msUAVs is

$$P_{r_max} = \max \{ P_t + G_t(\theta_1) - L_t + G_r(\theta_2) - L_r - L_p(d_1) \}, \quad (14)$$

where d_1 is the propagation distance between the GC and msUAVs.

To realize the effective interference of the AmsUAVs to the msUAVs uplink, the following inequation must be satisfied.

$$P_{t_AmsUAVs} + G_t(\theta_1) - L_t + G_r(\theta_2) - L_r - L_p(d_2) - Th_{msUAVs} \geq P_{r_max}, \quad (15)$$

where $P_{t_AmsUAVs}$ is the transmit power of the AmsUAVs, Th_{msUAVs} is the anti-jamming threshold of the msUAVs, d_2

is the propagation distance between the AmsUAVs and msUAVs. In general, d_2 is greater than d_1 .

Suppose the GC and AmsUAVs have the same $G_t(\theta_1)$ and L_t , then for a given d_2 , the interference power must satisfy

$$P_{t_AmsUAVs} \geq P_{r_max} - G_t(\theta_1) + L_t - G_r(\theta_2) + L_r + L_p(d_2) + Th_{msUAVs}. \quad (16)$$

3.5 Interference analysis

With the high-power noise emitted by the AmsUAVs, not only does the bottom noise in the surrounding airspace greatly increase but the main and side lobes of the antenna also cause interference to other nearby radio services, thus seriously affecting their normal operation.

The interference power received by the disturbed receiver is closely related to the operating frequency of the transmitted signal, the transmit power, and the distance between the transmitter and the receiver [36]. According to the INR criterion, measures that should be taken to avoid harmful interference include increasing the frequency separation, reducing the transmit power, increasing off-axis angles, or increasing the distance between transmitters and disturbed receivers [37, 38].

According to (2) and (5), when the AmsUAVs generates interference, the receiver of adjacent frequency bands can only work normally when the following equation is satisfied.

$$I - N = P_{t_AmsUAVs} + G_t(\theta_1) - L_t + G_r(\theta_3) - L_r - L_p(d_3) - FDR(\Delta f) - N_{bs} < \eta_{bs}, \quad (17)$$

where d_3 is the propagation distance between the AmsUAVs and base station; $G_r(\theta_3)$ is the antenna gain of the jammed receiver (e.g., TD-LTE) in the direction of

interference when the off-axis angle is θ_3 ; N_{bs} is mainly generated by the thermal motion of charged particles, and its power is

$$N_{bs} = -174 + 10 \log B + NF \quad \text{dBm}, \quad (18)$$

where B is the medium frequency bandwidth (Hz) and NF denotes the noise figure (dB).

In case when the radio wave propagation distance d_3 is given, the relationship between the transmit power of the AmsUAVs and frequency separation can be obtained as follows.

$$P_{t_AmsUAVs} < \eta_{bs} - G_t(\theta_1) + L_t - G_r(\theta_3) + L_r + L_p(d_3) + FDR(\Delta f) + N_{bs}. \quad (19)$$

Then, the maximum transmit power is $P_{t_max} = \max\{P_{t_AmsUAVs}\}$. According to (14), $P_{t_AmsUAVs} \in [P_{t_min}, P_{t_max}]$.

In case $P_{t_AmsUAVs}$ is given, according to (17), the frequency and distance separation between the AmsUAVs and the base station of mobile communication systems can be obtained as follows.

$$20 \log(d_3) \geq P_{t_AmsUAVs} + G_t(\theta_1) - L_t + G_r(\theta_3) - L_r - FDR(\Delta f) - N_{bs} - 32.4 - 20 \log f - \eta_{bs}. \quad (20)$$

Algorithm 1 provides a summary of the entire procedure.

4 Numeric simulation and discussion

In this section, we present the simulation results to verify the proposed interference analysis method. The simulation scenario includes a msUAVs, a GC, an AmsUAVs, a TD-LTE base station. DQPSK modulated signals are

Algorithm 1 Interference analysis algorithm steps

Input: Frequency f , bandwidth B , antenna gain G_t and G_r , the effective height h_1 and h_2 , the height of the base station H_b , noise figure NF , INR protection requirements η_{CDMA} and η_{TD-LTE} .

- 1: Calculate P_{r_max} , the maximum power received by the receiver of msUAVs.
- 2: Calculate the minimum equivalent transmission power $P_{t_AmsUAVs}$ of AmsUAVs according to the propagation distance between AmsUAVs and msUAVs.
- 3: Calculate the path loss L_p according to the radio wave propagation environment and propagation distance between AmsUAVs and the base station.
- 4: Calculate the maximum equivalent transmission power P_{t_max} of AmsUAVs according to the θ_2 , θ_3 , Δf , η_{CDMA} , η_{TD-LTE} and L_p between AmsUAVs and the base station.

Output: frequency separation Δf and maximum transmission power P_{t_max} .

transmitted by the msUAVs. The flight altitude of the msUAVs is less than 1000 m, and the simulation parameters of the msUAVs are shown in Table 1.

To determine the maximum power received by the msUAVs, the free space model is used as the propagation model to calculate the path loss. Figure 2 shows the signal power received by the msUAVs at different propagation distances and receiving off-axis angles.

When $\theta_2 = 0^\circ$, there is no off-axis angle between the receiving antenna and the main beam direction of the GC. $\theta_2 \neq 0^\circ$ indicates that the receiving antenna deviates from the main beam direction at θ_2 . When $\theta_2 = 0^\circ$, the power of the received signal is maximum; the larger the off-axis angle, the smaller the received signal power. For a given transmit power, the greater the transmission distance, the smaller the received signal power.

In practical applications, the effective propagation distance between GC and msUAVs varies randomly. For example, the height of msUAVs is relatively low, that is, h_2 relatively small. When the minimum received power is satisfied, the transmission power can be reduced adaptively. Assuming that the minimum received power of msUAVs is -50dBm , which can meet the minimum signal-to-noise ratio requirements, and $\theta_1 = 0^\circ$. Figure 3 shows comparison of the minimum transmit power of the GC for different propagation distances and off-axis angle θ_2 . Adaptively reducing the transmission power can reduce the energy consumption of GC and prolong the single charge service time of GC; it can also reduce the harmful interference of transmission power to other radio receivers.

Figure 4 shows comparison of the minimum interference power of the AmsUAVs for different propagation distances and off-axis angle θ_1 . To block the uplink communication system of the msUAVs, the minimum noise interference signal power transmitted by the AmsUAVs should be greater than or equal to the sum of the maximum received signal power and the anti-interference threshold of the msUAVs. In general, when the main beam direction of the AmsUAVs is directly opposite to the GC, i.e., $\theta_1 = 0^\circ$, the transmitted power is the least. However, to avoid interference with other radio communication systems in the same direction, the main beam direction can be deviated

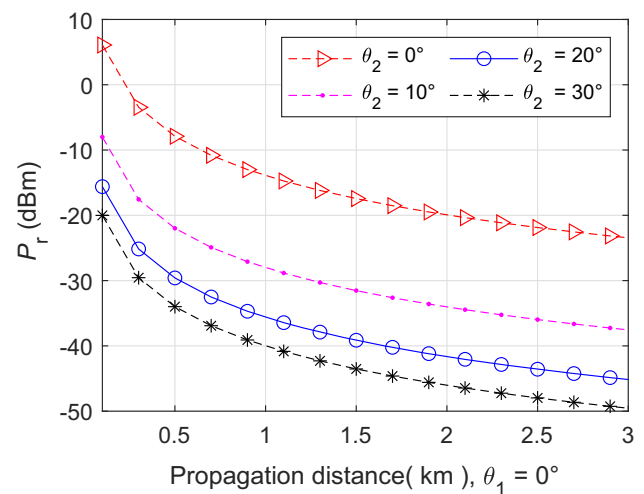


Fig. 2 Power of the received signal corresponding to different propagation distances, where $\theta_1 = 0^\circ$

from the GC, i.e., $\theta_1 \neq 0^\circ$. A short communication link between the AmsUAVs and msUAVs ensures less interference power, and thus a low possibility of interfering with other radio services. Considering the effect of path loss, the minimum interference power obtained by the free space radio propagation model is the least. With other radio propagation models, the minimum interference power will be larger.

A high-power noise-jamming signal transmitted by the AmsUAVs will interfere with radio equipment located within a certain distance. To avoid harmful interference, the AmsUAVs and base station need to meet certain frequency and distance separation as mentioned in Subsect. 3.5.

For AmsUAVs and TD-LTE base station, Figs. 5 and 6 show frequency separation versus distance separation for different off-axis angles in an urban scenario. The simulation parameters of TD-LTE base station are shown in Table 2. Due to the large bandwidth of TD-LTE, the center frequency interval between AmsUAVs and TD-LTE base stations is much larger than that between AmsUAVs and CDMA base stations [24]. When the central frequency separation is greater than 19.5MHz, distance separation obviously decreases. In practical application, the location

Table 1 Simulation parameters of msUAVs

	$B(\text{kHz})$	$P_t(\text{dBm})$	$G_{t,r}^1(\text{dBi})$	$h_{1,2}^2(\text{m})$	$L_{t,r}^3(\text{dB})$	$NF(\text{dB})$	$\eta_{\text{msUAVs}}(\text{dB})$
Transmitter	50	33	20	1.5	2	5	/
Receiver	50	/	3	<1000	1	5	-3

¹ $G_{t,r}$ means G_t or G_r , denote the antenna gain of the transmitter and the receiver respectively.

² $h_{1,2}$ means h_1 or h_2 , denote the effective height of the transmitter and the receiver respectively.

³ $L_{t,r}$ means L_t or L_r , denote the feeder loss of the transmitter and the receiver respectively

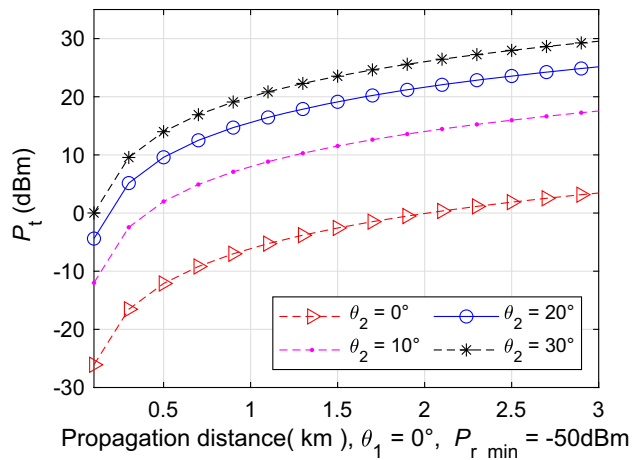


Fig. 3 Minimum transmit power corresponding to different propagation distances, where $\theta_1 = 0^\circ$ and the minimum received power $P_{r_min} = -50\text{dBm}$

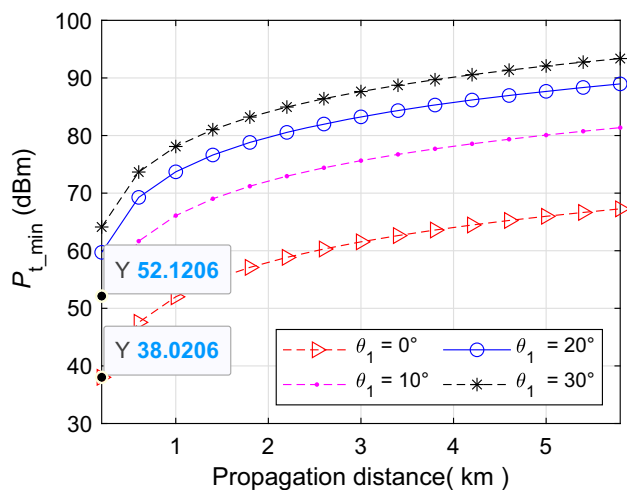


Fig. 4 Minimum interference power of the AmsUAVs corresponding to different propagation distances

of msUAVs is random, while the location of base station is fixed. In order to suppress or block the msUAVs uplink in a timely and effective manner, the location of AmsUAVs is also limited. AmsUAVs are also somewhat maneuverable and their position can be adjusted. Within a certain range, the off-axis angles between AmsUAVs and base station can be effectively increased by adjusting the position of AmsUAVs, so as to avoid harmful interference to base station. Similarly, increasing frequency separation is also an effective method to reduce interference. Comparing Figs. 5, 6, we can get that reducing the transmit power P_{t_min} can also effectively reduce the distance separation. Therefore, in practical applications, the transmit power of the AmsUAVs and off-axis angle should be reasonably selected.

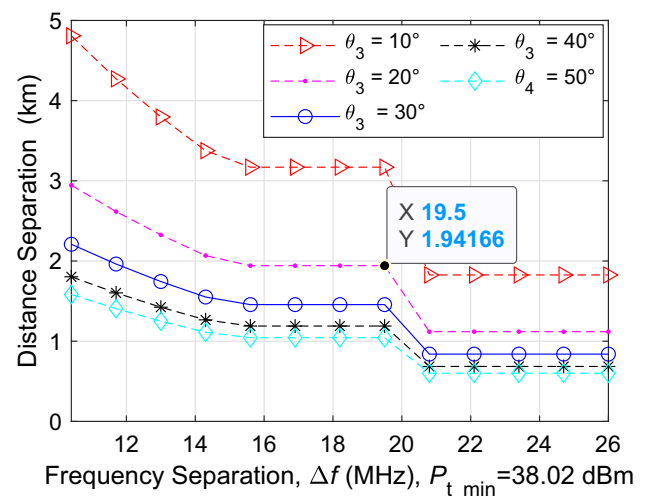


Fig. 5 Frequency separation with respect to distance separation under different off-axis angles for AmsUAVs and TD-LTE base station in an urban scenario with $P_{t_min} = 38.02\text{dBm}$

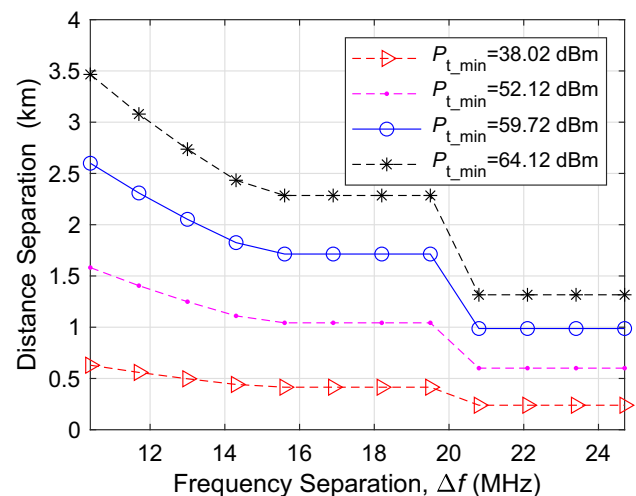


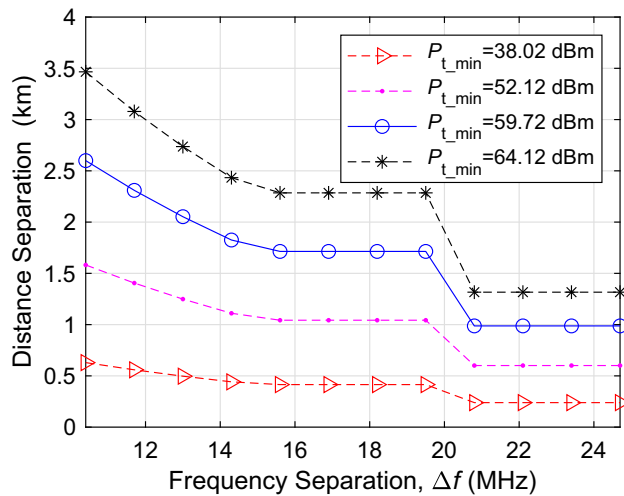
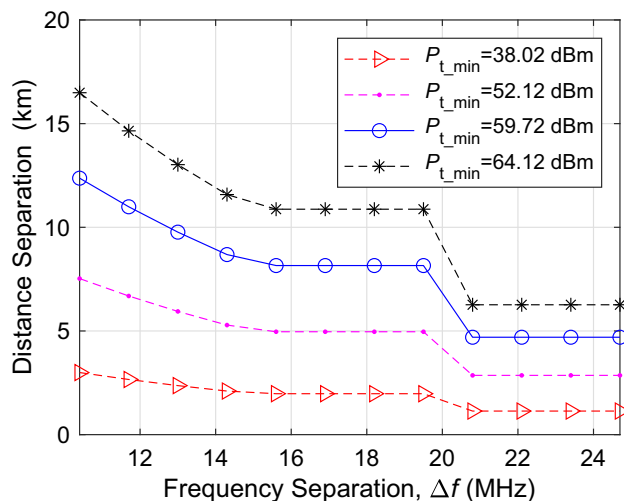
Fig. 6 Frequency separation with respect to distance separation under different off-axis angles for AmsUAVs and TD-LTE base station in an urban scenario with $P_{t_min} = 52.12\text{dBm}$

According to the radio propagation theory, path loss is an important factor in the interference analysis between different wireless communication systems. Figures 7 and 8 show frequency separation versus distance separation for different systems in suburban and open scenarios, respectively. Path loss is calculated using the Okumura–Hata suburban and the open environment propagation models, respectively. The figures indicate that the path loss of a suburban scenario is greater than that of an open scenario.

Therefore, under the condition of the same frequency separation, the distance separation of an open scenario is greater than that of a suburban scenario. Consequently, in an open scenario, the power of the interference signal should be carefully selected to avoid harmful interference

Table 2 Simulation parameters of the base station of mobile communication systems

	$B(\text{MHz})$	$G_r(\text{dBi})$	$H_b(\text{m})$	$L_r(\text{dB})$	$\text{NF}(\text{dB})$	$\eta_{\text{bs}}(\text{dB})$
TD-LTE Base station	20	15	30	2	5	-7

**Fig. 7** Frequency separation with respect to distance separation under different P_{t_min} for AmsUAVs and TD-LTE base station in a suburban scenario**Fig. 8** Frequency separation corresponding distance separation under different P_{t_min} for AmsUAVs and TD-LTE base station in an open scenario

to other radio systems. When the distance interval and frequency interval are constant, the main beam direction of the AmsUAVs should be adjusted within a certain range to increase the off-axis angle between the AmsUAVs and the base station, so as to effectively avoid harmful interference between the systems.

In practical application, in order to quickly and accurately block the connection between msUAVs and GC, AmsUAVs generally adopts the method of high-power suppression interference. In this method, the background noise is greatly raised by using the radio blocking and counteraction technology, so as to reduce the signal-to-noise ratio (SNR) of the receiver. According to the SNR protection requirements of msUAVs receiver, this method needs to calculate the minimum signal power and noise power that meet the SNR requirements. According to the requirements of short-term INR protection, the method proposed in this paper calculates the power of the minimum interference signal, which can adaptively obtain the minimum transmission power of AmsUAVs and minimize the interference of AmsUAVs to other radio communication systems.

5 Conclusion

In practice, the AmsUAVs not only interferes with msUAVs communication links but also can cause potentially harmful interference to civil aviation and mobile communication systems. This study analyzed the radio interference of the AmsUAVs and base station of the TD-LTE mobile communication systems. First, the minimum transmit power of the AmsUAVs in the corresponding frequency band was calculated on the basis of the parameters of the GC and the wave propagation distance. Second, according to the working frequency band, application scenario, and distance between AmsUAVs and base stations, interference analysis was performed on the basis of the INR criterion to obtain the frequency and distance separation that must be satisfied for both systems to coexist. Finally, the simulation verified the feasibility and effectiveness of the proposed method.

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Declarations

Conflict of interest/Conflict of interest All authors declare no conflict of interest.

Ethics approval This article does not contain any studies with human participants or animals performed by any of the authors.

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